

Management Information Systems

Phase 0: Process Architecture & Algorithmic Modeling

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1 Executive Summary of Selected Core Processes

In compliance with the Phase 0 requirements of the copper processing enterprise, five fundamental operational processes were identified, mapped, and mathematically structured for standard Business Process Model and Notation (BPMN 2.0) execution:

1. **Raw Material Transportation (Haulage Operations)**
2. **Geological Investigation & Blast Pattern Planning**
3. **Primary Comminution (Jaw Crushing Circuit)**
4. **Inventory Control & Supply Chain Buffer Optimization**
5. **Quality Control Assaying & Multi-Stage Grade Sampling**

2 Process 1: Raw Material Transportation (Haulage)

2.1 Architectural Swimlanes (Organizational Roles)

- **Mine Haul Truck Operator**
- **Mine Dispatch Operations Center**
- **Fleet Maintenance Team**
- **Safety & Environmental Officer**
- **Mine Operations Engineering Team**

2.2 Procedural Workflow Execution

Note: Ore loading takes place before Step 4; payload discharge takes place before Step 6.

1. Process Initiation (Start Event):

- *Input Data:* Fragmented Run-of-Mine (ROM) rock muck piles cleared post-blasting.
- *Initiating Event:* Haulage Dispatch Order received from the Mine Dispatch Center based on dynamic shift scheduling (modeled as a **BPMN Message Start Event**).

2. Haul Truck Dispatch & Dynamic Assignment (Dispatch Center):

- Fetch real-time telematics from the active truck fleet (GPS location, fuel reserves, payload hours, and preventive maintenance status).
- Match optimal truck capacity to shovel location.
- Transmit target route and pit loading coordinates to the selected driver's in-cab terminal.

Exclusive Gateway [Fleet Availability Assessment]:

- **Truck Available:** Assign route parameters and proceed to haul cycle.
- **No Truck Ready:** Issue system delay flag; re-queue allocation cycle.

3. Transit to Loading Face (Haul Truck Operator):

- Acknowledge route assignment via onboard telemetry.
- Verify pit road conditions, berm integrity, and haul-road safety regulations.
- Maneuver vehicle into loading pocket and signal positioning to the loading shovel.

4. Verification of Full Payload Clearance (Truck Operator & Loading System):

- Obtain electronic loading clearance from the shovel payload scale / loading supervisor.
- Validate loaded weight against Gross Vehicle Weight Rating (GVWR) to prevent under/overloading.
- Lock bed safety pins, lower rock covers, and confirm transit readiness.

Exclusive Gateway [Payload Verification]:

- **Payload Optimal:** Dispatch haul truck toward crushing dump pocket.
- **Payload Deficient / Excessive:** Reposition truck for payload trim or completion.

5. Haulage Cycle to Discharge Destination (Operator & Dispatch Center):

- Identify optimum transit corridor based on traffic congestion alerts.
- Traverse designated haul road adhering strictly to ramp speed caps.
- Continuously stream telematics to the Dispatch Center.

Exclusive Gateway [Haul Road Condition]:

- **Road Clear & Safe:** Maintain steady haul cycle to crusher bin.
- **Obstruction / Spill Detected:** Reroute haul truck via alternate haul ramp.

6. Tipping & Unload Clearance (Operator & Crusher Unloading System):

- Back into designated dumping bay under guidance of optical docking sensors.
- Tip dump box, discharging feed into the primary crusher intake hopper.
- Register tare weight on static scale; lower bed and confirm complete material evacuation.

Exclusive Gateway [Discharge Verification]:

- **Complete Evacuation:** Clear tipping pocket; reset empty haul cycle.
- **Hang-up / Incomplete Discharge:** Initiate hydraulic vibratory clearance.

7. Return Cycle to Loading Terminal (Haul Truck Operator):

- Receive return routing vector from the central dispatch terminal.
- Drive empty return circuit while monitoring engine oil pressure, coolant, and brake temperatures.

Exclusive Gateway [Vehicle Health Telemetry]:

- **Healthy Telemetry:** Return directly to pit for subsequent loading cycle.
- **Mechanical Fault Detected:** Divert immediately to the heavy equipment repair depot.

8. Process Termination (End Event):

- Haulage mission fulfilled; run-of-mine ore staged ready for primary comminution.

3 Process 2: Geological Investigation & Survey

3.1 Architectural Swimlanes (Organizational Roles)

- Geology Team
- Geology Data Center
- Mine Management
- Drilling Team

3.2 Procedural Workflow Execution

1. Process Initiation (Start Event):

- *Input Data:* Historic block models, geophysical maps, target cut-off grades, and environmental baseline bounds.
- *Initiating Event:* Formal Geological Survey Commission issued by Mine Management.

2. Field Data Acquisition (Geology Team):

- Extract outcrop surface lithology samples and trench samples.
- Process satellite multispectral imagery and airborne magnetic survey data.
- Log geographic coordinates and stratigraphy notes in field GIS units.

3. Data Analytics & Geotechnical Modeling (Geology Data Center):

- Ingest geochemical assays into geoscientific modeling systems.
- Run 3D spatial interpolation and deposit grade distribution algorithms.
- Establish lithological formations, fracture domains, and mineralized envelopes.

Exclusive Gateway [Geotechnical Data Validation]:

- **Data Statistically Valid:** Advance to reserve delineation.
- **Data Insufficient / Outliers:** Trigger secondary field sampling loop.

4. Exploration Target Delineation (Geology Team):

- Reconcile geostatistical models with standard industrial mineral classifications.
- Identify localized ore zones exhibiting high economic recovery potential.
- Formulate prioritized coordinates for delineation borehole drilling.

5. Preparation of Engineering Drill Plans (Geology Team):

- Generate digital drilling patterns with accurate northing, easting, and elevation collars.
- Design bench access routes, burden spacing, and anticipated penetration depths.

Exclusive Gateway [Plan Completeness Assessment]:

- **Plan Fully Validated:** Transmit blueprint package to Mine Management.
- **Geometrical / Topographic Conflicts:** Re-adjust layout coordinates.

6. Administrative Review & Authorization (Mine Management):

- Evaluate operational access routes, blasting proximity, and projected drilling capital expenditures.
- Formulate formal administrative drilling decision.

Exclusive Gateway [Management Approval]:

- **Authorized:** Forward approved pattern to the Drilling Production Team.
- **Rejected / Revisions Required:** Return dossier to Geology for alignment.

7. Technical Handover to Drilling Squad (Geology & Drilling Teams):

- Distribute geocoded digital patterns directly into drill rig positioning computers.
- Conduct inter-departmental technical and geotechnical hazard briefing.

8. Process Termination (End Event):

- Survey complete and stamped; field bench primed for rig mobilization and blasthole drilling.

4 Process 3: Primary Comminution (Jaw Crushing Circuit)

4.1 Architectural Swimlanes (Organizational Roles)

- Operations Control Center (OCC)
- Jaw Crusher Field Operator
- Crusher Vibrating Feed System
- SCADA Monitoring & Instrumentation System
- Crushed Material Transport System

4.2 Procedural Workflow Execution

1. Process Initiation (Start Event):

- *Input Data:* Bulk Run-of-Mine ore dumped into primary feed surge bin.
- *Initiating Event:* Scheduled machine initialization and pre-start calibration signal.

2. Mechanical Setpoint Configuration (Jaw Crusher Operator):

- Adjust the Closed Side Setting (CSS) hydraulic toggle to match target output sizing.
- Modulate variable-speed vibrating grizzly feeder parameters.
- Inspect lubrication delivery lines and toggle bearing fluid pressure.

Exclusive Gateway [Mechanical Readiness Audit]:

- **Parameters Verified:** Authorize automatic start sequence.
- **Anomalous Calibration:** Rectify hydraulic settings and re-test.

3. Continuous Comminution Operations (Feeder & Crusher Operator):

- Feed rocky material continuously into the jaw crushing chamber.
- Exert compressive fracture energy via the eccentric pitman swing jaw.
- Crush oversized boulders until rock gravity-drops past the discharge opening.

Exclusive Gateway [Crushing Chamber Integrity]:

- **Normal Operation:** Maintain steady-state rock comminution.
- **Tramp Metal / Jam Detected:** Automatic trip stop; dispatch field clearing crew.

4. Instrumentation Telemetry & Product Quality Audit (SCADA System):

- Monitor discharge particle gradation using ultrasonic conveyor profile scanners.
- Track drive motor current (amperage), flywheel vibration, and bearing temperatures.
- Generate warning alarms if uncrushed coarse rock fractions exceed allowable limits.

Exclusive Gateway [Granulometric Sizing Compliance]:

- **Size Within Thresholds (P_{80} compliant):** Advance to transfer conveyor.
- **Oversized Out-of-Spec:** Adjust jaw setting and divert oversize to secondary crushing circuit.

5. Transfer to Secondary Circuit (Material Transport System):

- Regulate mass feed rate across downstream heavy-duty conveyor belts.
- Maintain continuous tramp-iron magnetic separation and dust suppression sprays.
- Feed stream into secondary cone crushers or double-deck vibrating screens.

6. Process Termination (End Event):

- Run-of-mine bulk reduced to calibrated gravel; ready for secondary stage comminution.

5 Process 4: Inventory Control & Stock Optimization

5.1 Architectural Swimlanes (Organizational Roles)

- Warehouse Management
- Inventory Management System (IMS)
- Production Operations Management
- Supply Chain Management (SCM)
- Sales & Shipping Logistics Management

5.2 Procedural Workflow Execution

Context: Operates continuously over warehouse stock, addressing consumables (acids, grinding balls, reagents) and commercial outputs (copper concentrate and refined cathodes).

1. Process Initiation (Start Event):

- *Input Data:* Real-time on-hand inventory levels, safety stock thresholds, and forward production schedules.
- *Initiating Event:* Periodic stock audit trigger or automated reorder point alert.

2. Inventory Data Ingestion & Reorder Audit (IMS):

- Reconcile warehouse bin balances and finished goods lot registries.
- Compare physical inventory against upper and lower safety control limits.
- Generate variance and safety stock margin reports.

Exclusive Gateway [Inventory Variance State]:

- **Stock within Safe Limits:** Maintain passive scheduled monitoring.
- **Stockout Risk or Excessive Saturation:** Escalate corrective requisition.

3. Demand Forecasting & Consumption Analytics (IMS & Production Management):

- Compute daily consumption metrics for processing reagents (frothers, collector, sulfuric acid).
- Evaluate sales contracts, delivery backlogs, and global market price momentum.
- Run statistical time-series forecasts predicting 30-day inventory burn rates.

Exclusive Gateway [Operating Demand Alignment]:

- **Balanced Flow:** Sustain standard operational schedule.
- **Imbalance Observed:** Trigger purchase order or adjust production quotas.

4. Raw Material Procurement Interventions (SCM):

- Compare required raw input volumes against active vendor contract lead times.
- Evaluate alternate sourcing from secondary depots versus new supplier purchase orders.
- Generate formal Purchase Orders (PO) to approved reagent vendors.

Exclusive Gateway [Material Sufficiency Check]:

- **Sufficient Consumable Reserves:** Continue plant production schedule.
- **Critical Stock Depletion:** Expedite urgent procurement orders.

5. Finished Goods Dispatch Planning (Sales & Dispatch Logistics):

- Track warehouse cathode stock against commitments in active sales contracts.
- Reconfigure shipping schedules or expedite rail/truck freight to avoid yard saturation.
- Formulate sales schedules preventing warehouse congestion.

Exclusive Gateway [Cathode Inventory Velocity]:

- **Sales Dispatch Velocity Normal:** Maintain routine dispatch cadence.
- **Finished Inventory Surplus:** Accelerate outbound logistics campaigns.

6. Execution of Inventory Balances (Warehouse & Production Management):

- Throttling smelter casting rates or concentrator output to balance downstream holding zones.
- Reorganize internal warehouse space according to throughput priority.

7. Process Termination (End Event):

- Enterprise stock optimized; materials staged for Quality Control verification.

6 Process 5: Quality Sampling of Ore, Concentrate, and Cathode

6.1 Architectural Swimlanes (Organizational Roles)

- Quality Control Unit (QC)
- Mining Section
- Mineral Processing Section
- Finished Goods Warehouse
- Quality Management System (QMS)

6.2 Procedural Workflow Execution

1. Process Initiation (Start Event):

- *Input Data:* Sampling target node (ROM ore, slurry/concentrate, or finished cathodes), ASTM/ISO sampling protocols, and clean collection apparatus.
- *Initiating Event:* Receipt of automated sampling call from Mining, Processing, or Warehousing units (modeled as **Start Event**).

2. Sampling Plan Scheduling & Protocol Definition (QMS & QC Unit):

- Review the sampling mandate against statistical quality assurance models.
- Designate sampling times, composite increments, and extraction points.
- Assign certified QC personnel, sample seals, and laboratory containers.

Exclusive Gateway [Sampling Requisition Clearance]:

- **Protocol Approved:** Deploy sampling crew to designated plant node.
- **Methodology Non-compliant:** Revise sampling protocol parameters.

3. Run-of-Mine Ore Sampling Execution (Mining Section & QC Unit):

- Extract composite grab and core samples from drill cuttings across pit benches.

- Pack in barcoded tamper-evident moisture-proof bags for assay determination.

4. Flotation Concentrate Slurry Sampling (Processing Section & QC Unit):

- Collect composite pulp samples from automatic stream cross-cut samplers along conveyor lines.
- Measure slurry specific gravity, filter cake moisture percentage, and prepare test splits.
- Seal split samples and dispatch under chain-of-custody to analytical lab.

5. Refined Cathode Chemical Assaying Sample (Warehouse & QC Unit):

- Perform stratified random sampling across finished commercial cathode bundles.
- Extract specimen coupons/shavings for optical emission spectroscopy or ICP-MS analysis.
- Index coupon barcodes against batch heat numbers and warehouse storage bins.

6. Process Termination (End Event):

- Sample lots secured, verified, and transferred to the laboratory for analytical assaying.